

# Whole Numbers, Fractions & Percents

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## 1 · Connect to Prior Knowledge

This module is the mathematical foundation for everything that follows in BUS 123. Every formula you will build — markup percentages, payroll calculations, breakeven analysis, and loan math — starts with the skills you'll practice here: whole number operations, fraction-to-decimal conversion, and percent calculations. If you have taken a math course before, some of this will feel familiar; pay close attention to the **business context** and the **Excel approach**, which add layers that go beyond a standard algebra class. Our case study company for this module is **Meridian Advisory Group**, a wealth management firm whose day-to-day operations generate exactly the kinds of arithmetic problems we will solve.

## 2 · Core Concepts

### Part A — Whole Number Operations

The four arithmetic operations — addition, subtraction, multiplication, and division — are the building blocks of business math. In a business context, each operation maps to a specific type of question:

| Operation             | Business Meaning                        | Excel Formula          |
|-----------------------|-----------------------------------------|------------------------|
| <b>Addition</b>       | Summing revenue across accounts         | =SUM(B2:B10) or =B2+B3 |
| <b>Subtraction</b>    | Net income = Revenue – Expenses         | =B5 - C5               |
| <b>Multiplication</b> | Total fees = Accounts × Fee per account | =B3 * C3               |
| <b>Division</b>       | Per-client fee = Total ÷ Client count   | =B7 / C7               |

**Worked Example:** Meridian Advisory Group begins Q3 with 48 client accounts, adds 12 new clients, and loses 3. The management fee is \$4,200 per account per quarter. Quarter-end accounts:  $48 + 12 - 3 = 57$ . Total quarterly fees:  $57 \times \$4,200 = \$239,400$ . In Excel: =B2+B3-B4 for the account count, then =D5\*C2 for total fees, always referencing cells rather than typing numbers directly into the formula.

### ■ Unit Mismatch Warning

Before multiplying, check that both numbers use the same unit and time period. Multiplying a quarterly account count by an annual fee rate gives a meaningless result. Label every column header with its unit (e.g., 'Fee/Account/Quarter') to catch this error early.

## Part B — Fractions in Context

A fraction expresses a part of a whole: the numerator is the piece being measured, and the denominator is the total universe. In business, fractions appear as staff allocations, market share figures, and service-line proportions. Excel converts fractions to decimals automatically when you divide the numerator by the denominator; format the result cell as a percentage or fraction if needed.

**Worked Example:** Meridian's 6-person advisory team is split evenly: 2 advisors each in Tax, Estate Planning, and Investment Management. Each service line represents  $2/6 = 1/3 \approx 0.3333$  of total capacity. If Meridian hires 3 more Tax advisors, the Tax team grows to 5 of 9 total advisors:  $5/9 \approx 0.5556 = 55.6\%$ . Note that the denominator changed from 6 to 9 — you must use the new total in the denominator.

### ■ Tip: The Denominator Is the Whole Universe

When the total changes (e.g., new hires), update the denominator before recalculating. Using an outdated denominator is one of the most common fraction errors in business analysis.

## Part C — Percent Calculations

A percent is a fraction expressed out of 100. Business math involves three distinct percent calculations, each requiring a different formula structure. Confusing them is a frequent source of errors:

| Operation             | Excel Formula      | What It Calculates                                      |
|-----------------------|--------------------|---------------------------------------------------------|
| Case 1 — % of a Whole | =whole * rate      | What is 15% of \$239,400? → =239400*0.15                |
| Case 2 — % Change     | =(new - old) / old | Fees: \$1.2M → \$1.255M →<br>=(1255000-1200000)/1200000 |
| Case 3 — Ratio as %   | =part / whole      | 4 of 25 clients renewed → =4/25 (format as %)           |

**Worked Example — Forward and Backward:** Meridian earned \$1,200,000 in AUM fees last year and projects a 7.5% increase. Projected fees: =1200000\*(1+0.075) = **\$1,290,000**. Actual fees came in at \$1,255,000. Actual percent change: =(1,255,000 - 1,200,000) / 1,200,000 = **4.58%** — below the 7.5% target. In Excel, always use cell references: =(B5-B4)/B4. Format the result cell as Percentage — do not manually multiply by 100.

### ■ Common Mistake: % Change vs. % of Total

These are different formulas. '% change' compares a value to its earlier self: (New–Old)/Old. '% of total' compares a part to a whole at one point in time: Part/Whole. Ask yourself: am I measuring how something changed over time, or what share it represents of a bigger number?

## 3 · Formula Reference

| Operation               | Excel Formula      | What It Calculates            |
|-------------------------|--------------------|-------------------------------|
| <b>SUM range</b>        | =SUM(B2:B10)       | Add all values in a range     |
| <b>Arithmetic ops</b>   | =B2+B3-B4          | Add/subtract individual cells |
| <b>Multiply</b>         | =B3 * C3           | Qty × Price, Accounts × Fee   |
| <b>Divide</b>           | =B7 / C7           | Total ÷ Count = per-unit      |
| <b>Fraction→Decimal</b> | =B2 / C2           | Numerator ÷ Denominator       |
| <b>% of Whole</b>       | =whole * rate      | e.g. =B4 * 0.15               |
| <b>% Change</b>         | =(new - old) / old | Base = older number always    |
| <b>Ratio as %</b>       | =part / whole      | Format result as Percentage   |
| <b>Grow by rate</b>     | =B4 * (1 + rate)   | e.g. =B4*(1+0.075)            |

## 4 · Check Your Understanding

Answer each question before looking at the answer key on the following page. Show your Excel formula approach, not just the arithmetic result.

1. Meridian starts a quarter with 57 accounts and loses 4. What is the new account total? Write the Excel formula.
2. Total quarterly fee revenue was \$239,400. Tax services generated \$85,000 of that. What fraction — and what percent — of total fees came from Tax?
3. Meridian is targeting a 12% increase in total AUM fees from a base of \$239,400/quarter. What is the quarterly fee target? Write the Excel formula.
4. Actual fees came in at \$268,200. The prior quarter was \$239,400. What was the percent change? Which formula case does this represent?

5. Meridian's Investment team of 2 advisors represents what fraction of a 9-person total team? Express as a decimal rounded to 4 places.
6. True or False: the denominator in a percent change calculation should always be the newer (larger) number.
7. Meridian runs 4 service lines. Two lines each have 15 clients, one has 18 clients, one has 9 clients. What percent of total clients does the line with 18 clients represent?

## Answer Key · Check Your Understanding

- 1 53 accounts. Excel: =57-4 or =B2-B3. Unit check: both are counts of accounts.  
.
- 2 Fraction:  $85,000/239,400 \approx 0.3551$ . Percent:  $\approx 35.51\%$ . Excel: =85000/239400 → format as Percentage.  
This is Case 3 (ratio as %).
- 3 \$268,128. Excel: =239400\*(1+0.12) or =B4\*(1+0.12). This is Case 1 (% of whole, grow-by variant).  
.
- 4 Percent change =  $(268,200 - 239,400) / 239,400 \approx 12.03\%$ . Excel: =(268200-239400)/239400. This is Case 2 (% change). The base is the older number (\$239,400).  
.
- 5  $2/9 \approx 0.2222$ . Excel: =2/9. Note: the denominator is 9 (total team), not 6 (old team).  
.
- 6 False. The denominator in a percent change calculation should always be the **older (earlier)** number.  
.
- 7 18 clients out of  $(15+15+18+9) = 57$  total.  $18/57 \approx 31.58\%$ . Excel: =18/57 → format as Percentage. This is Case 3.  
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## 5 · Key Vocabulary

### Numerator

The top number in a fraction; represents the part being measured.

### Denominator

The bottom number in a fraction; represents the total or whole.

### Percent

A ratio expressed as a fraction out of 100. Written with the % symbol.

### Percent Change

A measure of how much a value has grown or declined relative to its starting point. Formula:  $(\text{New} - \text{Old}) / \text{Old}$ .

**Percent of Total**

A ratio comparing a part to the whole at a single point in time. Formula:  $\text{Part} / \text{Whole}$ .

**Cell Reference**

In Excel, using a cell address (e.g., B4) in a formula rather than typing a number directly, so the formula recalculates automatically when the input changes.

**Unit Mismatch**

An error that occurs when two values being multiplied or compared use different units or time periods (e.g., mixing annual and quarterly rates).

**AUM Fees**

Assets Under Management fees — revenue a financial advisory firm earns based on the total value of client assets it manages.